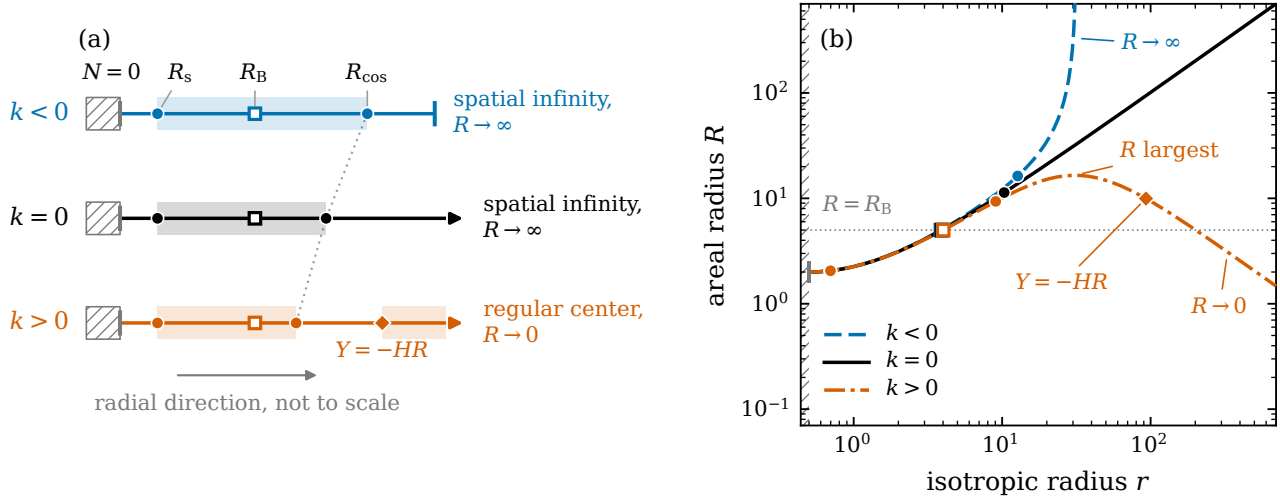
(a)
$N = 0$
R_s
R_B
R_{cos}
$k < 0$
spatial infinity,
$R \to \infty$
$k = 0$
spatial infinity,
$R \to \infty$
$k > 0$
regular center,
$R \to 0$
$Y = -HR$
radial direction, not to scale
(b)
$R \to \infty$
R largest
$R = R_B$
$Y = -HR$
$R \to 0$
$k < 0$
$k = 0$
$k > 0$
areal radius R
isotropic radius r
10^{-1}
10^0
10^1
10^2
lapse vanishes, $\mu = 1$
marginal spheres, $Y = HR$
marginal sphere, $Y = -HR$
cavity wall
untrapped, $\chi > 0$
antitrapped, $\chi < 0$
outside the region $N > 0$